

Enhancing Two-Photon Absorption of Green Fluorescent Protein by Quantum Entanglement

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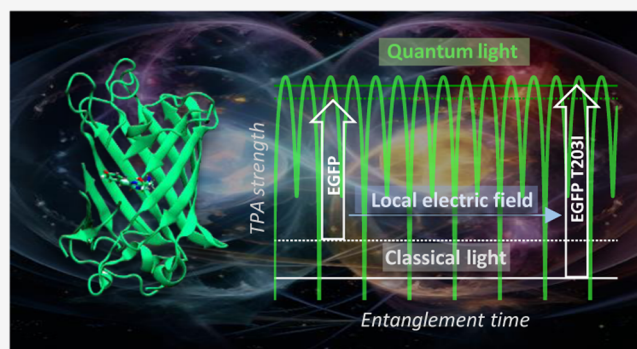


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Supporting Information

ABSTRACT: Exploring the electronic states of molecules through excitation with entangled and classical photon pairs offers new insights into the nature of light–matter interactions and stimulates the development of quantum spectroscopy. Here, we address the importance of temporal entanglement of light in two-photon absorption (TPA) upon the $S_0 \rightarrow S_1$ transition by the green fluorescent protein (GFP)—a key molecular unit in the bioimaging of living cells. By invoking a two-level model applicable when permanent dipole pathways dominate the two-photon transition, we derive a convenient closed-form analytical expression for the entangled TPA strength. For the first time, we disclose specific molecular properties that cause classical and entangled two-photon absorptions to be qualitatively different when exciting the same state. We reveal a new nonclassical contribution to the TPA strength, which is defined by the magnitude and directional alignment of permanent dipole moments in the initial and final states. Using high-level electronic structure theory, we show that the nonclassical contribution is intrinsically larger than the classical counterpart in GFP, leading to an enhancement of the TPA strength due to quantum entanglement by several orders of magnitude. We also present evidence that the classical and quantum TPA strengths can be modulated differently by the protein environment and demonstrate how to control the outcome by alterations in the local electric field of the protein caused by a single amino acid replacement. Our findings establish physical grounds for enhancing TPA in photoactive proteins by quantum entanglement, facilitating the rational design of high-efficiency biomarkers for future applications that utilize quantum light.



INTRODUCTION

Two-photon spectroscopy using entangled photon pairs is an actively developing area of research, being a subject of numerous theoretical^{1–13} and experimental^{14–21} works, as well as several reviews^{22–26} and tutorials.^{27,28} Quantum entanglement in a system of two (and more) particles is a specific correlation manifested in the fact that they behave in some sense like a single quantum object. For instance, the absorption rate of entangled photon pairs depends linearly on the flux density, whereas in classical two-photon absorption experiments, this dependence is quadratic.²⁹ This allows spectroscopic measurements to be carried out at significantly lower excitation light intensities (up to several orders of magnitude) than in classical experiments. This feature of absorption of entangled photon pairs can find its application in situations where there is a risk of photobleaching and other types of damage to samples, in particular when studying biological objects.^{15,21,30}

Although the effect of significant enhancement of two-photon absorption (TPA) in the case of entangled photons compared to classical two-photon absorption has been repeatedly mentioned in the literature, the expected magnitude of this enhancement, the influence of the specific structure of

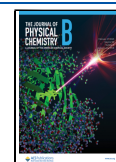
intermediate levels of chromophores, as well as the experimental setup required for reliable detection of the absorption of entangled photon pairs are still the subject of active discussions.^{9,11,31–38} Experience in theoretical prediction of entangled two-photon absorption (ETPA) cross sections of molecules is very limited and concerns diatomic (NO , HF , N_2)¹ as well as selected polyatomic chromophores, such as thiophene dendrimers,^{3,39} flavin mononucleotide,¹² and zinc tetraphenyl porphyrin.²⁰ Enhancement of two-photon absorption and selective excitation by entangled photon pairs are usually attributed to the difference between radiative lifetimes of entangled and unentangled two-photon excited states.^{3,20} Control over two-photon absorption can also be gained through optimization of entangled light, such as squeezing and pulse shaping.^{39,40} However, a fundamental question remains

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unanswered: Are there specific molecular properties of an absorber that cause classical and entangled two-photon absorptions to be qualitatively different when exciting the same state?

The green fluorescent protein (GFP)⁴¹ and its modifications, such as enhanced GFP (EGFP),⁴² are widely used in bioimaging and two-photon excitation microscopy.⁴³ Therefore, they represent ideal targets for analyzing the importance of quantum entanglement of light upon their two-photon excitation. At the same time, we are not aware of any theoretical work devoted to the absorption of entangled photon pairs by fluorescent proteins. Here, we perform a theoretical study on the absorption of temporally entangled photon pairs by EGFP and its single-point modification EGFP T203I (see Figure 1) upon the $S_0 \rightarrow S_1$ transition. For

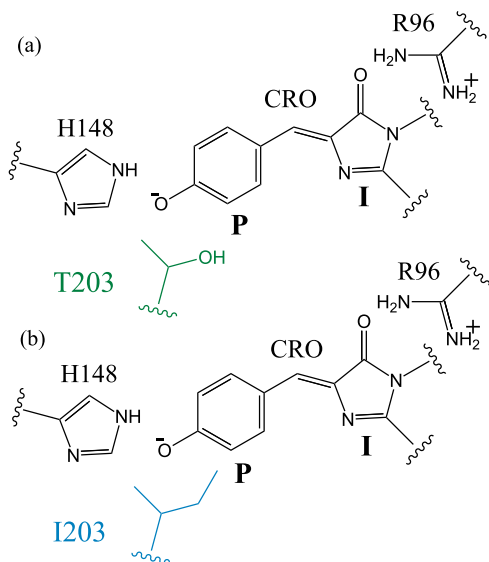


Figure 1. Chemical structures of the GFP chromophore anion and its immediate environment in (a) EGFP with polar threonine (T203, green) and (b) EGFP T203I with nonpolar isoleucine (I203, blue) at position 203. The two rings of the chromophore (CRO) are denoted as P and I. Note the zero total charge of the systems.

simplicity, we consider only vertical transitions between discrete electronic levels of the chromophores; hence, we neglect molecular vibrations and other effects leading to spectral broadening, in particular, a finite lifetime of excited states. We use the sum-over-states approach and adopt a simple two-level model (TLM), which has proven suitable for theoretical studies of two-photon transitions in a certain class of polar molecules.^{44–46} Under the assumptions presented above, we derive a closed-form analytical expression for the ETPA strength. We analyze ETPA of EGFP and EGFP T203I and reveal those molecular properties that determine their classical and entangled two-photon $S_0 \rightarrow S_1$ transition strengths. Our results disclose a new nonclassical contribution to ETPA that has a clear physical interpretation. In the GFP proteins, this contribution is significantly larger than that of the classical counterpart, leading to an enhancement of the two-photon transition strength by several orders of magnitude. Finally, we show how to control the outcome using the local electric field of the protein.

THEORY

The rate of entangled two-photon absorption (ETPA) is linear in the flux density of correlated photon pairs, and the absorption cross section is given by²⁹

$$\sigma_e \propto \frac{\omega_1 \omega_2}{A_e T_e} \delta_e g_e \quad (1)$$

where ω_1 and ω_2 are the photon energies; T_e is the entanglement time; A_e is the entanglement area of the photon pair; g_e is the line shape factor; and δ_e is the quantity we are primarily interested in, which we will call the ETPA transition strength and which will be defined below. After averaging over all orientations of absorbing molecules in the medium, the ETPA transition strength reads:

$$\delta_e = \frac{1}{30} \times \sum_{\alpha, \beta} (F \cdot S_e^{\alpha\alpha} S_e^{\beta\beta*} + G \cdot S_e^{\alpha\beta} S_e^{\alpha\beta*} + H \cdot S_e^{\alpha\beta} S_e^{\beta\alpha*}) \quad (2)$$

where $\alpha, \beta = x, y, z$; $S_e^{\alpha\beta}$ is the entangled two-photon transition moment; and F, G , and H are the known coefficients, which depend on the polarization of the two photons.⁴⁷ For the absorption of a degenerate entangled photon pair with $\omega_1 = \omega_2 = \omega$, we get

$$S_e^{\alpha\beta} = \sum_j \frac{\mu_{fj}^{\alpha} \mu_{j0}^{\beta} + \mu_{fj}^{\beta} \mu_{j0}^{\alpha}}{\Delta^{(j)}} (1 - e^{-i\Delta^{(j)} T_e}) \quad (3)$$

where $\mu_{ik}^{\gamma} = \langle \psi_i | \mu_{\gamma} | \psi_k \rangle$; μ^{γ} ($\gamma = x, y, z$) are the Cartesian components of the dipole moment operator; $|\psi_0\rangle, |\psi_j\rangle, |\psi_f\rangle$ denote the initial, final, and j -th intermediate states of the absorber; and E_0, E_j and E_f are the corresponding energy levels. The summation is carried out over all states of the absorber, including $j = 0$ and $j = f$.

$$\Delta^{(j)} = E_j - E_0 - \omega \quad (4)$$

is the detuning energy for the j -th intermediate state of the absorber. Under the two-photon resonance conditions, we fix the single photon energy as follows:

$$\omega = (E_f - E_0)/2 = \Omega_f/2 \quad (5)$$

where Ω_f is the two-photon resonance transition energy. We assume that intermediate states j exhibit relatively large lifetimes $1/\kappa_j$ and therefore neglect a decay term in eq 3, since typically $\kappa_j T_e \ll 1$.²⁹

An important issue concerns the predominant mechanism of the two-photon transition: either through a near-resonant intermediate state of the chromophore (intermediate state pathway, ISP) or through a process involving permanent dipole moments in the initial and final electronic states (permanent dipole pathway, PDP).^{1,48} The two possible PDPs correspond to two terms with $j = 0$ and $j = f$ in the sum of eq 3. It has been shown that the PDP channels can make the largest contribution to the classical TPA cross sections for polar chromophores if the two conditions are fulfilled simultaneously: (i) direct $0 \rightarrow f$ one-photon excitation is allowed; (ii) a permanent dipole moment changes substantially under this excitation.⁴⁹ In particular, the fluorescent proteins such as EGFP and its analogues meet these criteria for the $S_0 \rightarrow S_1$ transition.^{45,46} Accordingly, when only two terms with $j = 0$ and $j = f$ are considered in eq 3, we arrive at the so-called two-level model and obtain the following approximate closed-form

expression for the ETPA transition strength, δ_e (see the Supporting Information for the detailed derivation):

$$\delta_e = \bar{\delta}_e - 2\delta_c \cos((\Omega_f T_e)/2) - \frac{(\delta_+ - \delta_c)}{2} \cos(\Omega_f T_e) \quad (6)$$

where:

$$\delta_c = \frac{4|\mu_{ff} - \mu_{00}|^2 |\mu_{f0}|^2}{15\Omega_f^2} \times [(2F + G + H)\cos^2(\theta_-) + G + H] \quad (7)$$

is the classical TPA transition strength within the TLM, whereas

$$\delta_+ = \frac{4|\mu_{ff} + \mu_{00}|^2 |\mu_{f0}|^2}{15\Omega_f^2} \times [(2F + G + H)\cos^2(\theta_+) + G + H] \quad (8)$$

is the nonclassical contribution arising from entanglement of the two photons absorbed; $\theta_{\pm}$ are the angles between the vectors μ_{f0} and $(\mu_{ff} \pm \mu_{00})$; and

$$\bar{\delta}_e = \frac{\delta_+ + 3\delta_c}{2} \quad (9)$$

is the average value of δ_e over the period $4\pi/\Omega_f$.

Equations 6, 8, and 9 constitute the main result of this work, and to the best of our knowledge, none of them have been previously reported in the literature. For the special case of orthogonally polarized photons obtained through the collinear type II spontaneous parametric down-conversion (SPDC) process,⁵⁰ the polarization-dependent coefficients F , G , and H in eq 2 are equal to⁴⁷

$$F = H = -1; \quad G = 4 \quad (10)$$

The same coefficients F , G , and H are used here for calculating both the classical δ_c and entangled δ_e TPA transition strengths. As a result, eqs 7 and 8 are rewritten as follows:

$$\delta_c = \frac{4|\mu_{ff} - \mu_{00}|^2 |\mu_{f0}|^2}{15\Omega_f^2} (\cos^2(\theta_-) + 3) \quad (11)$$

$$\delta_+ = \frac{4|\mu_{ff} + \mu_{00}|^2 |\mu_{f0}|^2}{15\Omega_f^2} (\cos^2(\theta_+) + 3) \quad (12)$$

COMPUTATIONAL DETAILS

Vertical electronic transition energies, permanent and transition dipole moments of the proteins to be used in eqs 6, 11, and 12, are calculated using the hybrid quantum mechanical/molecular mechanical (QM/MM) approach, which is based on a high-level electronic structure theory method—the second-order extended multiconfiguration quasi-degenerate perturbation theory (XMCQDPT2).⁵¹ The details of the computational protocol have been published previously.⁴⁶ In brief, the full atomistic models of the EGFP and EGFP T203I proteins in the ground electronic state are obtained based on the X-ray structure (PDB ID: 1EMA)⁵² using molecular dynamics simulations with annealing followed by PBE0/(aug)-cc-pVDZ//CHARMM geometry optimization. The excited-state calculations are performed at the XMCQDPT2[7]/SA(7)-CASSCF(14,13)/(aug)-cc-pVDZ level of theory by employing

the electrostatic embedding QM/MM scheme. The calculations are performed using the Firefly quantum chemistry package.⁵³

RESULTS AND DISCUSSION

We illustrate the application of the derived equations by calculating the ETPA strength for the $S_0 \rightarrow S_1$ transition in EGFP and its modified variant, EGFP T203I. In the modified version, the polar threonine amino acid residue close to the chromophore is replaced by nonpolar isoleucine at position 203 (Figure 1). In the case of the $S_0 \rightarrow S_1$ transition, the final state index f equals 1, and the key parameters of the present ETPA model are shown in Figure 2. The calculated vertical

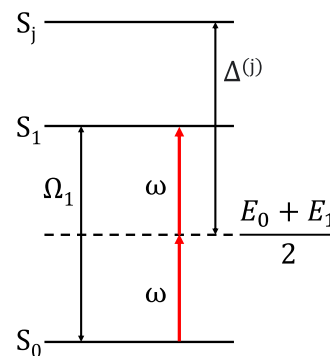


Figure 2. Energy diagram displaying the key parameters of the ETPA model for the $S_0 \rightarrow S_1$ transition.

transition energy, along with the permanent and transition dipole moments, as well as the classical and nonclassical contributions to the ETPA strength, are presented in Table 1 (for an extended data set, see Table S1).

Table 1. Permanent and Transition Dipole Moments, Half of the Two-Photon Resonance Transition Energy, Classical δ_c and Nonclassical δ_+ Contributions to the ETPA Strength of the $S_0 \rightarrow S_1$ Transition for EGFP and EGFP T203I (in Atomic Units)

	EGFP	EGFP T203I
$ \mu_{00} $	13.22	13.97
$ \mu_{11} $	12.43	13.51
$ \mu_{11} - \mu_{00} $	1.75	0.98
$ \mu_{11} + \mu_{00} $	25.60	27.47
$ \mu_{10} $	4.11	4.29
$\Omega_1/2$	4.62×10^{-2}	4.46×10^{-2}
θ_-, deg	4	11
θ_+, deg	119	122
δ_c	6.44×10^3	2.35×10^3
δ_+	1.12×10^6	1.52×10^6

The calculated ETPA strength δ_e as a function of entanglement time T_e for EGFP and EGFP T203I is shown in Figure 3. The results of the TLM model, which accounts for the PDP transition mechanism only, are here compared to those obtained using the sum-over-states approach according to eqs 2, 3, and 10 with up to 7 lowest electronic states, $j = 0, \dots, 6$. It follows that the use of the TLM is justified in this case, and the PDP channels make the predominant contribution to the overall ETPA strength. The same conclusion has been

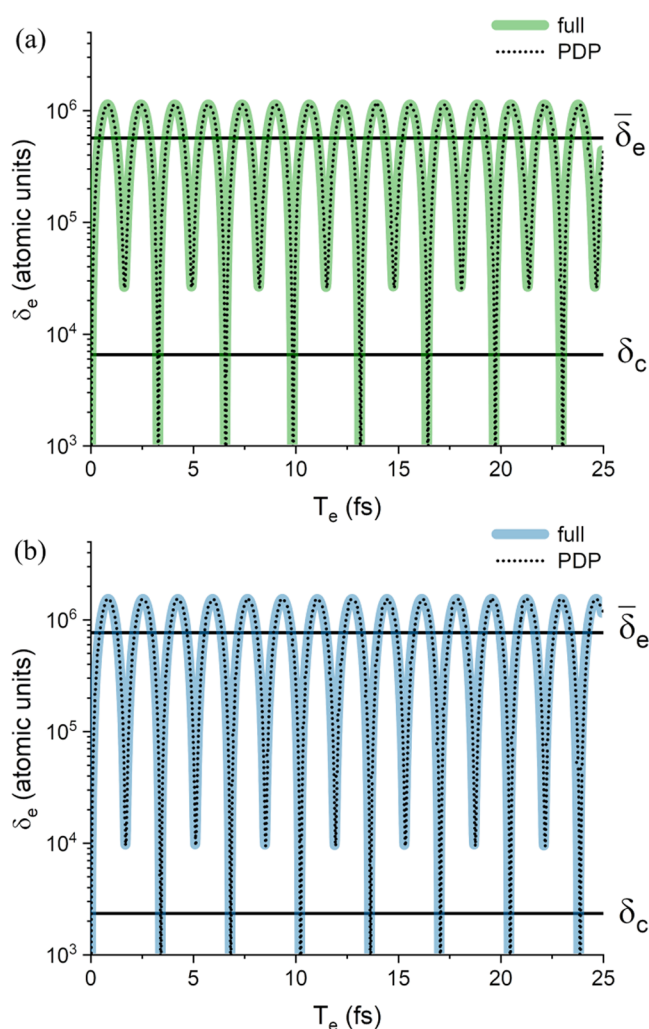


Figure 3. ETPA strength δ_e as a function of entanglement time T_e for the $S_0 \rightarrow S_1$ two-photon transition in (a) EGFP and (b) EGFP T203I: with only permanent dipole pathways included (PDP, eq 6) and from the direct sum-over-states calculations according to eqs 2 and 3 (full); see the text for further details. The horizontal lines denote classical TPA value δ_c and average ETPA value $\bar{\delta}_e$. Since we neglect a finite lifetime of excited states, there is no decay of the ETPA strength as entanglement time increases.

made earlier based on the theoretical analysis of the classical TPA values for the same proteins.⁴⁶

It follows from eq 6 that δ_e is a periodic function of T_e with two frequencies, Ω_1 and $\Omega_1/2$, and the corresponding periods of 1.65 and 3.3 fs, respectively. This is the reason for the alternating maxima and minima observed in Figure 3. Physically, they can be interpreted as constructive and destructive interference of the two permanent dipole pathways, $0 \rightarrow 0 \rightarrow 1$ and $0 \rightarrow 1 \rightarrow 1$, respectively, mentioned above. The deep minima of δ_e go to zero, when $\cos(\Omega_1 T_e) = \cos((\Omega_1 T_e)/2) = 1$. Following Fei et al.,²⁹ these zeros can be associated with full entanglement-induced two-photon transparency. The shallower minima at $\delta_e = 4\delta_c$ occur when $\cos(\Omega_1 T_e) = 1$ and $\cos((\Omega_1 T_e)/2) = -1$ simultaneously. The maxima of δ_e are reached when $\cos(\Omega_1 T_e) = -1$ (and hence $\cos((\Omega_1 T_e)/2) = 0$) and equal to $\delta_c + \delta_+$ (see eq 6). Since $\delta_+ \gg \delta_c$ for both EGFP and EGFP T203I (see Table 1), we conclude that the maximum gain in their two-photon

absorptivity induced by entanglement is primarily determined by the value of δ_+ (eq 12).

The prevailing PDP mechanism for the $S_0 \rightarrow S_1$ two-photon transition enables a lucid interpretation of the nonclassical contribution to the ETPA strength, which is directly related to the properties of the absorbing molecule. The classical and nonclassical terms both depend on the permanent dipole moments; however, the nonclassical contribution depends on their vector sum Σ_μ , whereas the classical term is defined by their difference Δ_μ . The directions of these vectors are similar in both proteins and are illustrated in Figure 4.

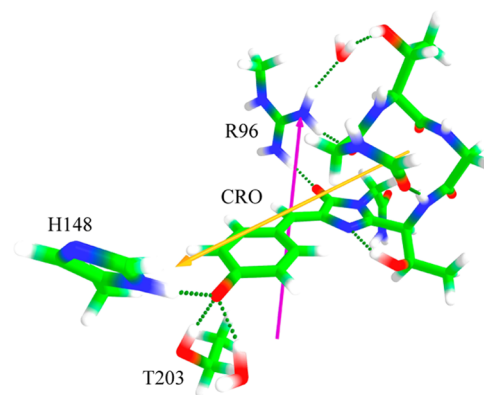


Figure 4. Directions of the linear combinations of the permanent dipole moments of the initial and final states in EGFP: Σ_μ (violet) and Δ_μ (orange). Note that the arrows go through the XYZ origin and only represent the direction and not the length of the vectors. The chromophore (CRO) is negatively charged, while the arginine residue (R96) has a positive charge. Shown is only the QM part of the protein. The C, O, N, and H atoms are depicted in green, red, blue, and white, respectively.

Upon the lowest-energy $\pi\pi^*$ excitation, the major changes occur in the plane of the chromophore along the line connecting its two rings, phenolate (P) and imidazolinone (I) (see Figure 1), which is collinear to the transition dipole moment, μ_{10} . As expected, μ_{11} and μ_{00} as well as their sum, Σ_μ , are directed from negatively charged P to positively charged R96, which is bound to I through a hydrogen bond (see Figure 4). Although these vectors have other components, their projections onto μ_{10} vary the most depending on the state and local electric field of the protein. The projection of Σ_μ is large and directed from P to I. The Δ_μ vector is nearly collinear to the direction of μ_{10} . Since $|\mu_{00}| > |\mu_{11}|$ (see Table 1), Δ_μ is directed from I to P, which is opposite to the direction of charge transfer upon excitation. In EGFP and EGFP T203I, the sum is much larger than the corresponding difference due to nearly parallel vectors μ_{00} and μ_{11} , and the projections of Δ_μ and Σ_μ onto μ_{10} turn out to be antiparallel in both proteins (see Figure S1). As a result, the ETPA strength of these proteins is intrinsically higher than that of classical two-photon absorption. Additionally, changes of the local electric field within the protein are expected to modulate the nonclassical and classical contributions differently.

By replacing the polar T203 amino acid residue with its nonpolar I203 analogue in EGFP T203I, the local electric field close to the chromophore inside the protein alters. The electric field induced by the dipole of the $O^{\delta-}H^{\delta+}$ group of T203 is directed from I to P, and its projection onto μ_{10} turns out to be parallel to Δ_μ and antiparallel to Σ_μ . Since the change in the

permanent dipole moment of the initial and final states induced by the field, and hence the corresponding polarizability, is larger in S_1 , Δ_μ decreases while Σ_μ increases in EGFP T203I (see Table 1 and Figure S1). Therefore, the single-point modification of EGFP demonstrates opposite impacts on the two-photon absorption of entangled and classical photon pairs. This modification enhances ETPA while reducing classical TPA in EGFP T203I, which lacks the additional electric field of the hydroxyl group. The overall enhancement of the two-photon absorption due to quantum entanglement turns out to be 4 times greater in EGFP T203I compared to that in EGFP.

For the first time, we show that in the GFP proteins upon their $S_0 \rightarrow S_1$ two-photon transition, the nonclassical contribution is intrinsically larger than the classical counterpart, leading to the enhancement of the TPA strength due to quantum entanglement by several orders of magnitude. We also present evidence that the classical and quantum TPA strengths can be modulated differently by the protein environment, thereby providing proof of our concept: There are specific molecular properties that cause classical and entangled two-photon absorptions to be qualitatively different when exciting the same state. An intrinsic enhancement of TPA by several orders of magnitude through quantum entanglement of light is anticipated to be a common feature of all photoactive proteins with charged (polar) chromophores when the permanent dipole moments of the initial and final states are nearly aligned and point in the same direction. Increasing the local electric field of a protein in this direction will further enhance ETPA. The varying effects of this field on entangled versus classical TPA will be observed when the permanent dipole moment of the chromophore is larger in the initial state, while its polarizability is larger in the final state.

CONCLUSIONS

Our findings provide physical grounds for the enhancement of two-photon absorption in fluorescent proteins induced by quantum entanglement. By deriving closed-form analytical expressions, we show that the transition strengths for two-photon excitation of the same nondegenerate final state are qualitatively different when using classical versus entangled photon pairs. The difference is due to the nonclassical contribution to ETPA, which is defined by the sum of the permanent dipole moments of the initial and final states. We also demonstrate how to control the entanglement-induced enhancement of two-photon absorption in proteins using a local electric field. The equations derived in this work are relevant to a wide range of fluorescent proteins, photoreceptors, and various other chromophores and dyes that satisfy the conditions for applying the two-level model. The $S_0 \rightarrow S_1$ transition, which is optically allowed as one-photon excitation in these polar systems, and directionally aligned permanent dipole moments of the initial and final states make them perfect candidates for experimental studies on entangled two-photon absorption. Our results pave the way for the rational design of high-efficiency ETPA biomarkers for future fluorescent microscopy applications that utilize quantum light.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.4c07869>.

Extended quantum chemistry data, detailed derivation of the equations for entangled two-photon transition strengths within the framework of the two-level model, and graphical representations of permanent dipole moments and their linear combinations (PDF)

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Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Burdick, R. K.; Varnavski, O.; Molina, A.; Upton, L.; Zimmerman, P.; Goodson, T. I. Predicting and Controlling Entangled Two-Photon Absorption in Diatomic Molecules. *J. Phys. Chem. A* **2018**, *122*, 8198–8212.
- (2) León-Montiel, R. J.; Svozilik, J.; Torres, J. P.; U'Ren, A. B. Temperature-Controlled Entangled-Photon Absorption Spectroscopy. *Phys. Rev. Lett.* **2019**, *123*, No. 023601.
- (3) Kang, G.; Avnaki, K. N.; Mosquera, M. A.; Burdick, R. K.; Villabona-Monsalve, J. P.; Goodson, T. I.; Schatz, G. C. Efficient Modeling of Organic Chromophores for Entangled Two-Photon Absorption. *J. Am. Chem. Soc.* **2020**, *142*, 10446–10458.
- (4) Gu, B.; Keefer, D.; Aleotti, F.; Nenov, A.; Garavelli, M.; Mukamel, S. Photoisomerization transition state manipulation by entangled two-photon absorption. *Proc. Natl. Acad. Sci. U.S.A.* **2021**, *118*, No. e2116868118.
- (5) Li, B.; Hofmann, H. F. Energy-time entanglement and intermediate-state dynamics in two-photon absorption. *Phys. Rev. A* **2021**, *104*, No. 022434.
- (6) Schlawin, F. Polarization-Entangled Two-Photon Absorption in Inhomogeneously Broadened Ensembles. *Front. Phys.* **2022**, *10*, No. 848624.
- (7) Asban, S.; Chernyak, V. Y.; Mukamel, S. Nonlinear quantum interferometric spectroscopy with entangled photon pairs. *J. Chem. Phys.* **2022**, *156*, No. 094202.
- (8) Raymer, M. G.; Landes, T. Theory of two-photon absorption with broadband squeezed vacuum. *Phys. Rev. A* **2022**, *106*, No. 013717.
- (9) Drago, C.; Sipe, J. E. Aspects of two-photon absorption of squeezed light: The continuous-wave limit. *Phys. Rev. A* **2022**, *106*, No. 023115.

- (10) Fu, M.; Tabakaev, D.; Thew, R. T.; Wesolowski, T. A. Fine-Tuning of Entangled Two-Photon Absorption by Controlling the One-Photon Absorption Properties of the Chromophore. *J. Phys. Chem. Lett.* **2023**, *14*, 2613–2619.
- (11) Li, B.; Hofmann, H. F. Enhancement of broadband entangled two-photon absorption by resonant spectral phase flips. *Phys. Rev. A* **2023**, *108*, No. 013706.
- (12) Wittkop, M.; Marmolejo-Tejada, J. M.; Mosquera, M. A. Multichromatic quantum superpositions in entangled two-photon absorption spectroscopy. *Org. Electron.* **2023**, *120*, No. 106858.
- (13) Schlawin, F. Two-photon absorption cross sections of pulsed entangled beams. *J. Chem. Phys.* **2024**, *160*, No. 144117.
- (14) Varnavski, O.; Pinsky, B.; Goodson, T. I. Entangled Photon Excited Fluorescence in Organic Materials: An Ultrafast Coincidence Detector. *J. Phys. Chem. Lett.* **2017**, *8*, 388–393.
- (15) Villabona-Monsalve, J. P.; Varnavski, O.; Palfey, B. A.; Goodson, T. I. Two-Photon Excitation of Flavins and Flavoproteins with Classical and Quantum Light. *J. Am. Chem. Soc.* **2018**, *140*, 14562–14566.
- (16) Varnavski, O.; Goodson, T. I. Two-Photon Fluorescence Microscopy at Extremely Low Excitation Intensity: The Power of Quantum Correlations. *J. Am. Chem. Soc.* **2020**, *142*, 12966–12975.
- (17) Burdick, R. K.; Schatz, G. C.; Goodson, T. I. Enhancing Entangled Two-Photon Absorption for Picosecond Quantum Spectroscopy. *J. Am. Chem. Soc.* **2021**, *143*, 16930–16934.
- (18) Tabakaev, D.; Montagnese, M.; Haack, G.; Bonacina, L.; Wolf, J.-P.; Zbinden, H.; Thew, R. T. Energy-time-entangled two-photon molecular absorption. *Phys. Rev. A* **2021**, *103*, No. 033701.
- (19) Tabakaev, D.; Djorović, A.; La Volpe, L.; Gaulier, G.; Ghosh, S.; Bonacina, L.; Wolf, J.-P.; Zbinden, H.; Thew, R. T. Spatial Properties of Entangled Two-Photon Absorption. *Phys. Rev. Lett.* **2022**, *129*, No. 183601.
- (20) Varnavski, O.; Giri, S. K.; Chiang, T.-M.; Zeman, C. J.; Schatz, G. C.; Goodson, T. Colors of entangled two-photon absorption. *Proc. Natl. Acad. Sci. U.S.A.* **2023**, *120*, No. e2307719120.
- (21) Spennato, D.; Leone, J.; Gundhardt, C.; Varnavski, O.; Fabbri, R.; Caprini, M.; Zamboni, R.; Benfenati, V.; Goodson, T. I. Investigations of Astrocyte Calcium Signaling and Imaging with Classical and Nonclassical Light. *J. Phys. Chem. B* **2024**, *128*, 7966–7977.
- (22) Dorfman, K. E.; Schlawin, F.; Mukamel, S. Nonlinear optical signals and spectroscopy with quantum light. *Rev. Mod. Phys.* **2016**, *88*, No. 045008.
- (23) Schlawin, F.; Dorfman, K. E.; Mukamel, S. Entangled Two-Photon Absorption Spectroscopy. *Acc. Chem. Res.* **2018**, *51*, 2207–2214.
- (24) Basset, M. G.; Setzpfandt, F.; Steinlechner, F.; Beckert, E.; Pertsch, T.; Gräfe, M. Perspectives for Applications of Quantum Imaging. *Laser Photonics Rev.* **2019**, *13*, No. 1900097.
- (25) Szoke, S.; Liu, H.; Hickam, B. P.; He, M.; Cushing, S. K. Entangled light-matter interactions and spectroscopy. *J. Mater. Chem. C* **2020**, *8*, 10732–10741.
- (26) Zheltikov, A. M.; Scully, M. O. Photon entanglement for life-science imaging: rethinking the limits of the possible. *Phys.-Usp.* **2020**, *63*, 698.
- (27) Raymer, M. G.; Landes, T.; Marcus, A. H. Entangled two-photon absorption by atoms and molecules: A quantum optics tutorial. *J. Chem. Phys.* **2021**, *155*, No. 081501.
- (28) Zhang, Z.; You, C.; na Loaiza, O. S. M.; Fickler, R.; de J. León-Montiel, R.; Torres, J. P.; Humble, T. S.; Liu, S.; Xia, Y.; Zhuang, Q. Entanglement-based quantum information technology: a tutorial. *Adv. Opt. Photonics* **2024**, *16*, 60–162.
- (29) Fei, H.-B.; Jost, B. M.; Popescu, S.; Saleh, B. E. A.; Teich, M. C. Entanglement-Induced Two-Photon Transparency. *Phys. Rev. Lett.* **1997**, *78*, 1679–1682.
- (30) Ma, Y.-Z.; Doughty, B. Nonlinear Optical Microscopy with Ultralow Quantum Light. *J. Phys. Chem. A* **2021**, *125*, 8765–8776.
- (31) Landes, T.; Raymer, M. G.; Allgaier, M.; Merkouché, S.; Smith, B. J.; Marcus, A. H. Quantifying the enhancement of two-photon absorption due to spectral-temporal entanglement. *Opt. Express* **2021**, *29*, 20022–20033.
- (32) Lerch, S.; Stefanov, A. Experimental requirements for entangled two-photon spectroscopy. *J. Chem. Phys.* **2021**, *155*, No. 064201.
- (33) Raymer, M. G.; Landes, T.; Allgaier, M.; Merkouché, S.; Smith, B. J.; Marcus, A. H. How large is the quantum enhancement of two-photon absorption by time-frequency entanglement of photon pairs? *Optica* **2021**, *8*, 757–758.
- (34) Parzuchowski, K. M.; Mikhaylov, A.; Mazurek, M. D.; Wilson, R. N.; Lum, D. J.; Gerrits, T.; Camp, C. H.; Stevens, M. J.; Jimenez, R. Setting Bounds on Entangled Two-Photon Absorption Cross Sections in Common Fluorophores. *Phys. Rev. Appl.* **2021**, *15*, No. 044012.
- (35) Corona-Aquino, S.; Calderón-Losada, O.; Li-Gómez, M. Y.; Cruz-Ramírez, H.; Venicio, V. Á.; Carreón-Castro, M. P.; de León-Montiel, R.; U'Ren, A. B. Experimental Study of the Validity of Entangled Two-Photon Absorption Measurements in Organic Compounds. *J. Phys. Chem. A* **2022**, *126*, 2185–2195.
- (36) Triana-Arango, F.; Ramos-Ortiz, G.; Ramírez-Alarcón, R. Spectral Considerations of Entangled Two-Photon Absorption Effects in Hong-Ou-Mandel Interference Experiments. *J. Phys. Chem. A* **2023**, *127*, 2608–2617.
- (37) Martínez-Tapia, Á.; Corona-Aquino, S.; Triana-Arango, F.; You, C.; Jin, R.-B.; Magaña-Loaiza, O. S.; Dong, S.-H.; U'Ren, A. B.; León-Montiel, R. d. J. Witnessing entangled two-photon absorption via quantum interferometry. *APL Photonics* **2023**, *8*, No. 036104.
- (38) Landes, T.; Smith, B. J.; Raymer, M. G. Limitations in fluorescence-detected entangled two-photon-absorption experiments: Exploring the low- to high-gain squeezing regimes. *Phys. Rev. A* **2024**, *110*, No. 033708.
- (39) Giri, S. K.; Schatz, G. C. Manipulating Two-Photon Absorption of Molecules through Efficient Optimization of Entangled Light. *J. Phys. Chem. Lett.* **2022**, *13*, 10140–10146.
- (40) Oka, H. Efficient selective two-photon excitation by tailored quantum-correlated photons. *Phys. Rev. A* **2010**, *81*, No. 063819.
- (41) Tsien, R. Y. The Green Fluorescent Protein. *Annu. Rev. Biochem.* **1998**, *67*, 509–544.
- (42) Yang, T.-T.; Cheng, L.; Kain, S. R. Optimized Codon Usage and Chromophore Mutations Provide Enhanced Sensitivity with the Green Fluorescent Protein. *Nucleic Acids Res.* **1996**, *24*, 4592–4593.
- (43) Zipfel, W. R.; Williams, R. M.; Webb, W. W. Nonlinear magic: multiphoton microscopy in the biosciences. *Nat. Biotechnol.* **2003**, *21*, 1369–1377.
- (44) Drobizhev, M.; Makarov, N. S.; Tillo, S. E.; Hughes, T. E.; Rebane, A. Describing Two-Photon Absorptivity of Fluorescent Proteins with a New Vibronic Coupling Mechanism. *J. Phys. Chem. B* **2012**, *116*, 1736–1744.
- (45) Drobizhev, M.; Callis, P.; Nifosi, R.; Wicks, G.; Stoltzfus, C.; Barnett, L.; Hughes, T.; Sullivan, P.; Rebane, A. Long- and Short-Range Electrostatic Fields in GFP Mutants: Implications for Spectral Tuning. *Sci. Rep.* **2015**, *5*, No. 13223.
- (46) Aslopovsky, V. R.; Scherbinin, A. V.; Kleshchina, N. N.; Bochenkova, A. V. Impact of the Protein Environment on Two-Photon Absorption Cross-Sections of the GFP Chromophore Anion Resolved at the XMCQDPT2 Level of Theory. *Int. J. Mol. Sci.* **2023**, *24*, No. 11266.
- (47) Monson, P. R.; McClain, W. M. Polarization Dependence of the Two-Photon Absorption of Tumbling Molecules with Application to Liquid 1-Chloronaphthalene and Benzene. *J. Chem. Phys.* **1970**, *53*, 29–37.
- (48) Upton, L.; Harpham, M.; Suzer, O.; Richter, M.; Mukamel, S.; Goodson, T. I. Optically Excited Entangled States in Organic Molecules Illuminate the Dark. *J. Phys. Chem. Lett.* **2013**, *4*, 2046–2052.
- (49) Dick, B.; Hohlneicher, G. Importance of initial and final states as intermediate states in two-photon spectroscopy of polar molecules. *J. Chem. Phys.* **1982**, *76*, 5755–5760.
- (50) Sergienko, A. V.; Shih, Y. H.; Rubin, M. H. Experimental evaluation of a two-photon wave packet in type-II parametric downconversion. *J. Opt. Soc. Am. B* **1995**, *12*, 859–862.

(51) Granovsky, A. A. Extended multi-configuration quasi-degenerate perturbation theory: The new approach to multi-state multi-reference perturbation theory. *J. Chem. Phys.* **2011**, *134*, No. 214113.

(52) Ormö, M.; Cubitt, A. B.; Kallio, K.; Gross, L. A.; Tsien, R. Y.; Remington, S. J. Crystal Structure of the *Aequorea victoria* Green Fluorescent Protein. *Science* **1996**, *273*, 1392–1395.

(53) Granovsky, A. A. Firefly Version 8.2.0. <http://classic.chem.msu.su/gran/firefly> (accessed March 12, 2024).